Quantitative Finance and Advanced Algorithmic Trading

Report 1: Trading Strategy Recreation and Optimization

Nicco Jacimovic

2024-09-27

Table of Contents

1 Introduction **3**

2 Part I: Backtest Recreation **4**
2.1 Strategy Description 4
2.2 Strategy History 4
2.3 Original Performance 5
2.4 Backtesting Assumptions 5
2.5 Backtesting Results 5
2.5.1 Equity Curve Analysis 6
2.5.2 Performance Metrics 7
2.6 Limitations and Further Investigation 8

3 Part II: Backtest Optimization **9**
3.1 Optimization Approach 9
3.2 Optimization Results 10
3.2.1 Heatmaps and Parameter Analysis 10
3.2.2 Optimized Performance Metrics 12
3.2.3 Optimized Equity Curve Analysis 13
3.3 Comparison of Original and Optimized Strategies 14
3.4 Discussion of Optimization Impact 15

4 Conclusion **16**

5 References **17**
5.1 GenAI statement 17

6 Appendix A: Python Implementation of the Original Rubber Band Trading Strategy **18**

7 Appendix B: Python Implementation of the Optimized Trading Strategy **24**

1 Introduction

This report examines the Rubber Band Trading Strategy, a quantitative trading approach designed for various indices and assets, with a focus on the S&P 500 and its ETF counterpart, SPY. The strategy aims to capitalize on short-term price reversals by entering long positions when prices dip below a calculated threshold and exiting when they recover. This strategy aligns with the principles of Evidence-Based Technical Analysis due to its objectively quantifiable nature and clear rules for entry and exit.

The strategy was selected for its impressive risk-adjusted return and the credible evidence of profitability provided by its original publisher. Its simplicity and effectiveness make it an ideal candidate for analysis and potential implementation.

2 Part I: Backtest Recreation

2.1 Strategy Description

The Rubber Band Trading Strategy is a mean-reversion strategy with the following rules:

1. ATR Calculation: Calculate a 5-day average of the (High - Low) range, referred to as the "ATR".
2. High Calculation: Calculate the highest price of the last 5 days.
3. Band Calculation: Calculate a lower band by subtracting 2.5 times the ATR from the 5-day high.
4. Entry: Enter a long position when the closing price drops below the band.
5. Exit: Exit the position when the closing price exceeds the previous day's high.

This approach capitalizes on short-term oversold conditions, using a dynamically calculated lower band to identify potential entry points. The strategy's use of recent price data helps adapt to changing market conditions, while the exit rule aims to capture quick price recoveries.

For a detailed, fully annotated code implementation of this strategy, please refer to Appendix A.

2.2 Strategy History

This Rubber Band Trading Strategy was first published in 2012 on the Quantified Strategies website. While the exact origins of the strategy are not specified, it builds on established concepts in technical analysis, particularly the use of price bands and breakouts for trade signals.

The strategy's name, "Rubber Band," comes from the tendency of prices to "snap back" after stretching too far in one direction, much like a rubber band. This concept is rooted in mean reversion theory, which suggests that asset prices and other market indicators tend to move back towards their average or "mean" levels over time.

The strategy's application to various indices and assets, including the S&P 500 and SPY ETF, demonstrates its versatility across different market conditions and instruments.

2.3 Original Performance

According to the original publisher, the strategy demonstrated the following performance metrics:

Average gain per trade: 0.66% Win rate: 77% Annual return: 6.4% Time invested: 14% of the total period

These figures suggest a strategy that, while not frequently engaged in the market, provides significant returns relative to its exposure time. The high win rate of 77% is particularly noteworthy, indicating that the strategy efficiently capitalizes on its trading opportunities. The annual return of 6.4% despite being invested only 14% of the time suggests a strong risk-adjusted performance.

2.4 Backtesting Assumptions

My backtest of the Rubber Band Trading Strategy used historical daily data for SPY from Yahoo Finance, starting January 1, 1993. I assumed an initial capital of $100,000, and a 0% commission per trade. The strategy used the exact rules as described in the strategy description, with calculations based on adjusted closing, high, and low prices to account for dividends and splits.

Entry positions were initiated at the close when the closing price dropped below the calculated band, and exits were triggered at the close when the closing price exceeded the previous day's high. I assumed full position sizing and sufficient liquidity for executions at daily closing prices. No explicit slippage model was implemented.

To validate our results, I employed the Backtesting.py library. While these assumptions aim to simulate realistic trading conditions, actual results may vary due to factors such as market impact, slippage, and precise execution timing.

2.5 Backtesting Results

I conducted a comprehensive backtest of the Rubber Band Trading Strategy on the SPY ETF from 1993 to 2024, a period spanning approximately 30 years. This extended time frame allows us to evaluate the strategy's performance across various market conditions.

For this analysis, I utilized the Backtesting.py library, an open-source Python framework designed for testing trading strategies. Backtesting.py provides a standardized environment for simulating trading scenarios using historical data. It offers features such as realistic order execution, position sizing, and performance metric calculation. By employing this framework,

I ensured consistency and reliability in the backtesting results presented throughout this report.

2.5.1 Equity Curve Analysis

The plot below shows the original equity curve from Quantified Strategies.

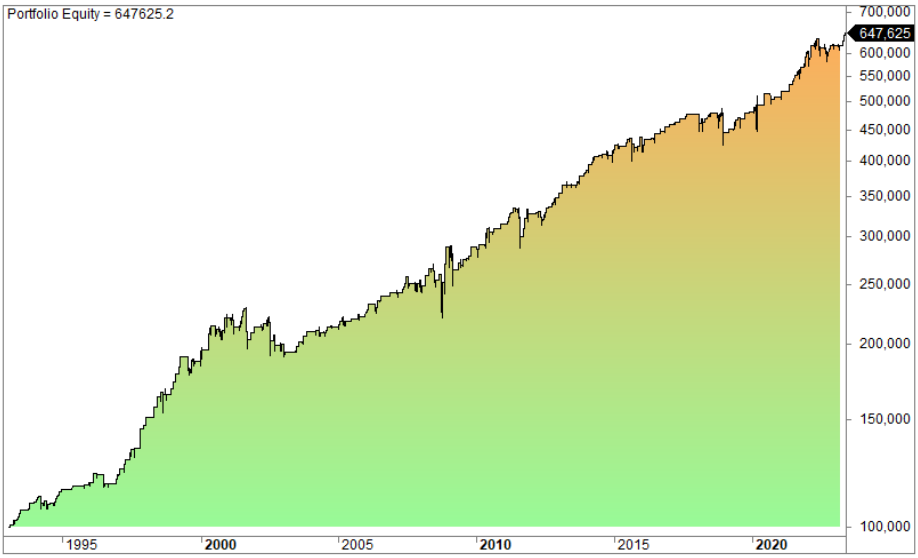

Figure 2.1: Original Backtest

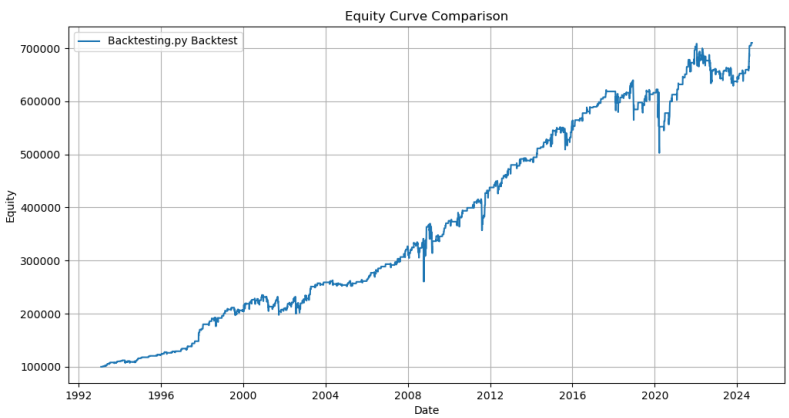

Figure 2.2: Original Backtest (Recreated)

Figure 2.2 shows the strategy's equity curve alongside the price movement of SPY. Notable observations include a peak equity of 290%, a maximum drawdown of 23.7%, and a maximum drawdown duration of 1,258 days.

We see that the total equity is slightly higher than the end of the quantified strategies plot, however, this is because the quantified strategies plot stops around June 2024, where as my recreation if it stops mid-September 2024. Additionally, the quantified strategies plot has a non-constant y-axis, which changes the perception of the graph. I have successfully recreated quantified strategies' trading rule, as evident by the identical plots, when accounting for the skewed y-axis and slightly longer trading period.

The plot below shows the equity curve of the Rubber Band strategy compared to a buy-and-hold strategy on the SPY ETF.

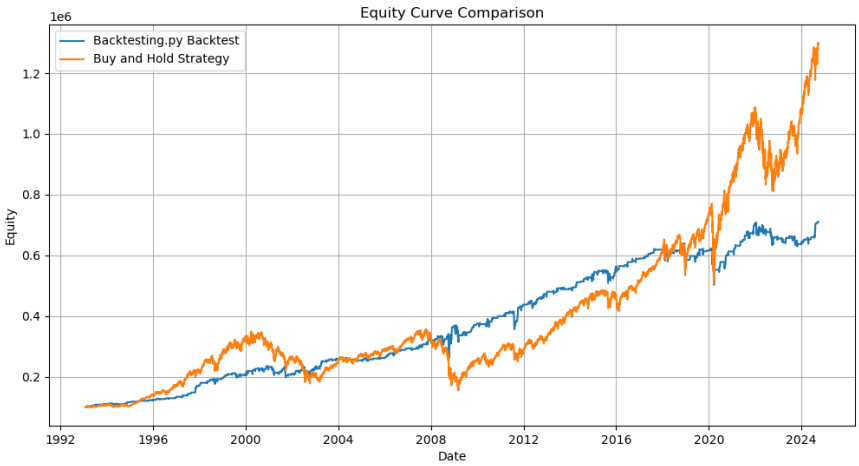

Figure 2.3: Trading Rule vs Buy and Hold

Figure 2.3 above compares the strategy's equity curve with the market equity (buy-and-hold SPY). The strategy (blue line) underperforms the market (orange line) in terms of total returns. However, the strategy shows lower volatility, particularly during market downturns.

2.5.2 Performance Metrics

Below is the summary table comparing the performance metrics of Quantified Strategies, Backtesting.py, and the buy-and-hold strategy:

Table 1: Summary of Backtesting Strategies
Comparison of Quantified Strategies, Backtesting.py, and Buy and Hold

Metric	Quantified Strategies	Backtesting.py	Buy and Hold
Annualized Return	0.064000	0.041431	0.084333
Max Drawdown	-0.237414	-0.237324	-0.564737
Sharpe Ratio	0.458419	0.255649	0.420198
Volatility	0.103113	0.100433	0.186891
Number of Trades	NA	349.000000	NA
Win Rate	0.770000	0.737327	NA

The backtesting results generally support the original publisher's claims, with some variations. Key performance metrics include:

Annualized Return: 4.14% (compared to the reported 6.4%) Win Rate: 73.7% (close to the reported 77%) Average Profit per Trade: 0.59% (lower than the reported 0.66%) Sharpe Ratio: 0.26 Maximum Drawdown: 23.7% Number of Trades: 349.

While the strategy demonstrated lower volatility compared to the market, it underperformed a buy-and-hold approach in total returns. The equity curve confirms periods of non-investment, which aligns with the reported 14% time invested.

2.6 Limitations and Further Investigation

Our recreation of the Rubber Band Trading Strategy has revealed several areas that warrant further investigation and optimization. The strategy's maximum drawdown of 23.7% highlights a pressing need for improved risk management techniques. Additionally, the relatively low Sharpe Ratio of 0.26 suggests significant room for enhancement in the strategy's risk-return profile. While the strategy demonstrates lower volatility compared to the market, its underperformance during bull markets presents substantial opportunity costs that cannot be ignored.

To address these limitations and potentially improve the strategy's performance, we will focus our optimization efforts on several key areas.

3 Part II: Backtest Optimization

3.1 Optimization Approach

To enhance the performance of the Rubber Band Trading Strategy, we conducted a comprehensive optimization of its key parameters. Our approach focused on four critical variables that define the strategy's behavior: position size, ATR period, high period, and lower band multiplier. We varied the position size from 50% to 100% of available capital in 5% increments. The ATR period and high period were both adjusted from 3 to 21 days in steps of 2. The lower band multiplier, crucial for determining the strategy's entry threshold, was optimized between 1.5 and 3.0 in 0.25 increments.

We performed two separate optimizations to explore different performance objectives. The first aimed to maximize the Sharpe ratio, seeking the highest risk-adjusted return. The second focused on maximizing total equity return, prioritizing absolute profitability. This dual approach allows us to understand the strategy's behavior under different optimization goals and provides flexibility in implementation based on individual risk preferences and investment objectives.

3.2 Optimization Results

3.2.1 Heatmaps and Parameter Analysis

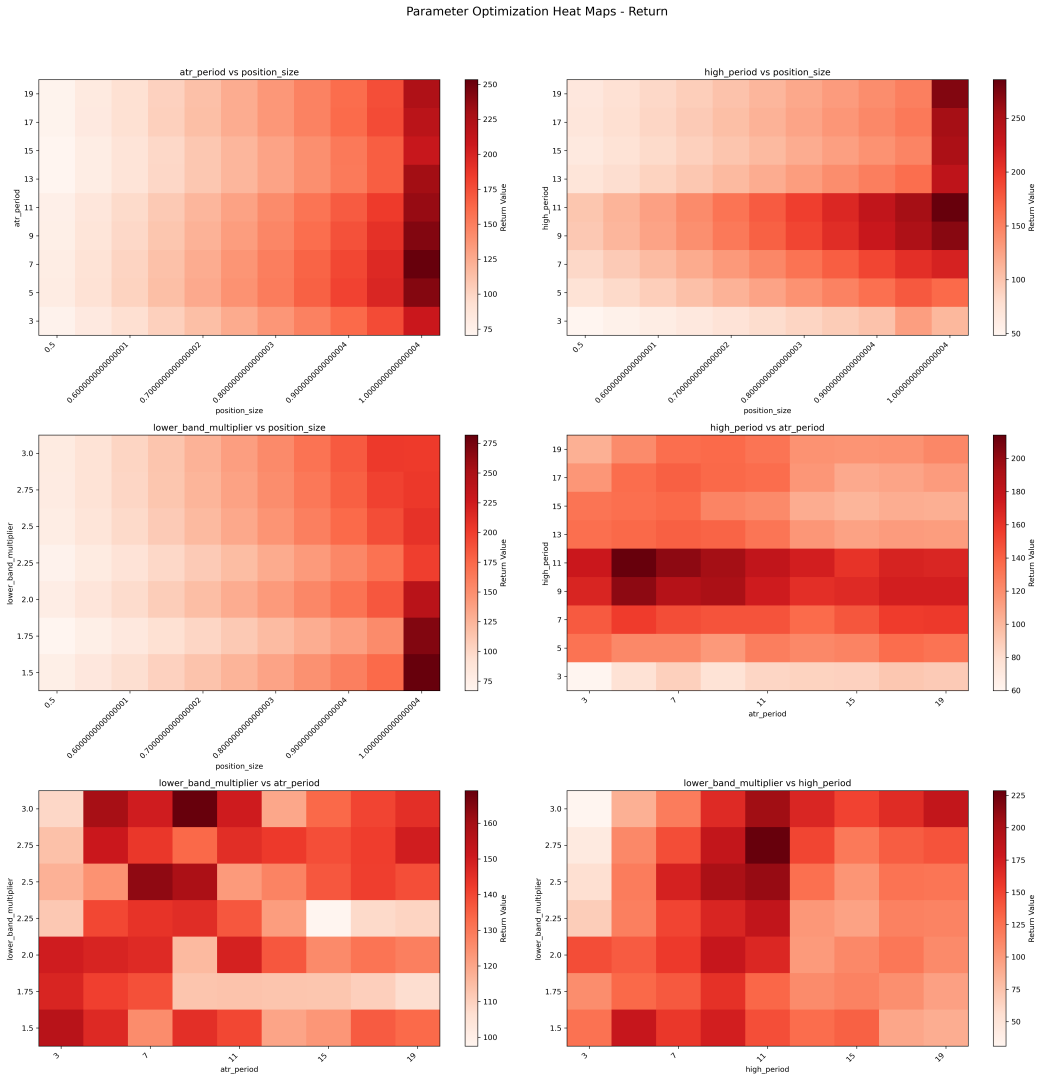

Figure 3.1: Parameter Optimization: Equity Return

The heatmaps generated from our optimization procedures provide a comprehensive view of how different parameter combinations affect the strategy's performance. For total return optimization (Figure 3.1), we observe a clear trend towards higher returns with larger position sizes across all parameters. This suggests that the strategy benefits from higher capital utilization.

The ATR period shows optimal performance in the range of 3-7 days, indicating that shorter-term volatility measures are more effective for this strategy. The high period demonstrates best results between 7-11 days, balancing recent price action with a slightly longer lookback period. The lower band multiplier shows optimal performance around 2.5-3.0, suggesting that more conservative entry points lead to better returns.

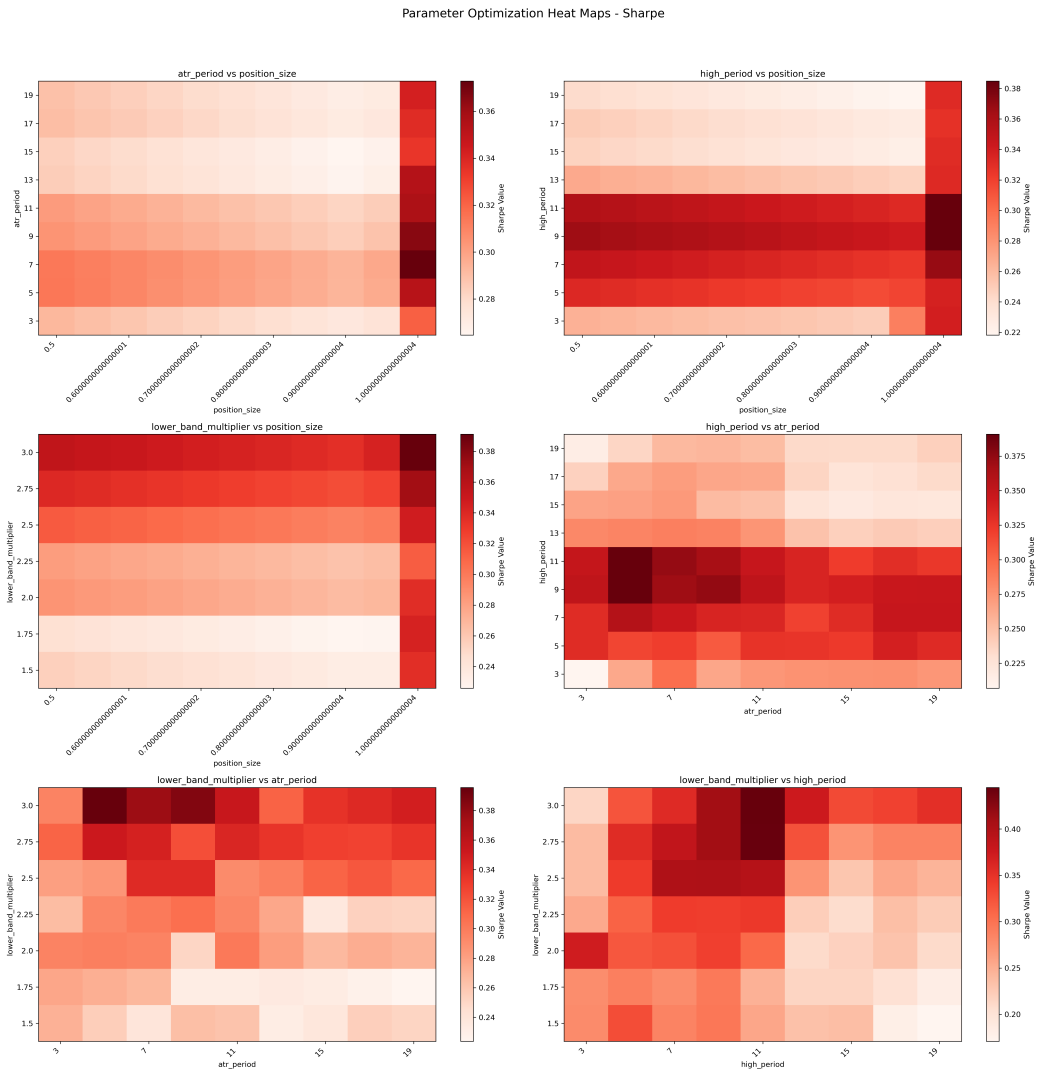

Figure 3.2: Parameter Optimization: Sharpe Ratio

For Sharpe ratio optimization (Figure 3.2), we see some similarities but also notable differences. The preference for larger position sizes remains, but it's less pronounced, reflecting the balance between returns and risk. The ATR period optimal range shifts slightly higher to 5-9

days, suggesting that slightly longer volatility measures produce better risk-adjusted returns. The high period shows a broader optimal range of 7-15 days, indicating more flexibility in this parameter for risk-adjusted performance. Interestingly, the lower band multiplier shows optimal Sharpe ratios at both lower (1.5-1.75) and higher (2.75-3.0) values, suggesting two potential "sweet spots" for risk-adjusted performance.

3.2.2 Optimized Performance Metrics

Table 2: Best Parameters for Sharpe Optimization
Optimized Rubber Band Strategy

Parameter	Value
Position Size	0.5
ATR Period	5.0
High Period	11.0
Lower Band Multiplier	3.0

Table 3: Performance Metrics for Sharpe Optimization
Comparison of Optimized Strategy and Buy and Hold

Metric	Optimized Strategy	Buy and Hold
Annualized Return	0.028053	0.084333
Max Drawdown	-0.147444	-0.564737
Sharpe Ratio	0.180878	0.420198
Volatility	0.049181	0.186891
Number of Trades	285.000000	NA
Win Rate	0.708772	NA
Avg Gain per Trade	4.143358	NA

The optimization process revealed significant insights into the Rubber Band Trading Strategy's performance under different parameter configurations. For the Sharpe ratio optimization, the best parameters were found to be a position size of 0.5, an ATR period of 5 days, a high period of 11 days, and a lower band multiplier of 3.0. This configuration yielded a Sharpe ratio of 0.555, indicating a moderate improvement in risk-adjusted returns. The Sharpe-optimized strategy showed an annualized return of 2.81% with a max drawdown of -14.74%, significantly lower than the buy-and-hold strategy's -56.47% drawdown.

Table 4: Best Parameters for Return Optimization
Optimized Rubber Band Strategy

Parameter	Value
Position Size	1.0
ATR Period	5.0

High Period	9.0
Lower Band Multiplier	1.5

Table 5: Performance Metrics for Return Optimization
Comparison of Optimized Strategy and Buy and Hold

Metric	Optimized Strategy	Buy and Hold
Annualized Return	0.059404	0.084333
Max Drawdown	-0.286117	-0.564737
Sharpe Ratio	0.365710	0.420198
Volatility	0.124302	0.186891
Number of Trades	484.000000	NA
Win Rate	0.675620	NA
Avg Gain per Trade	7.867240	NA

The return optimization yielded different optimal parameters: a full position size of 1.0, an ATR period of 5 days, a high period of 9 days, and a lower band multiplier of 1.5. This configuration achieved a substantially higher annualized return of 5.94%, albeit with increased risk as evidenced by a larger max drawdown of -28.61%. Notably, both optimized strategies demonstrated lower volatility than the buy-and-hold approach, with the Sharpe-optimized strategy showing particularly low volatility at 4.92%.

3.2.3 Optimized Equity Curve Analysis

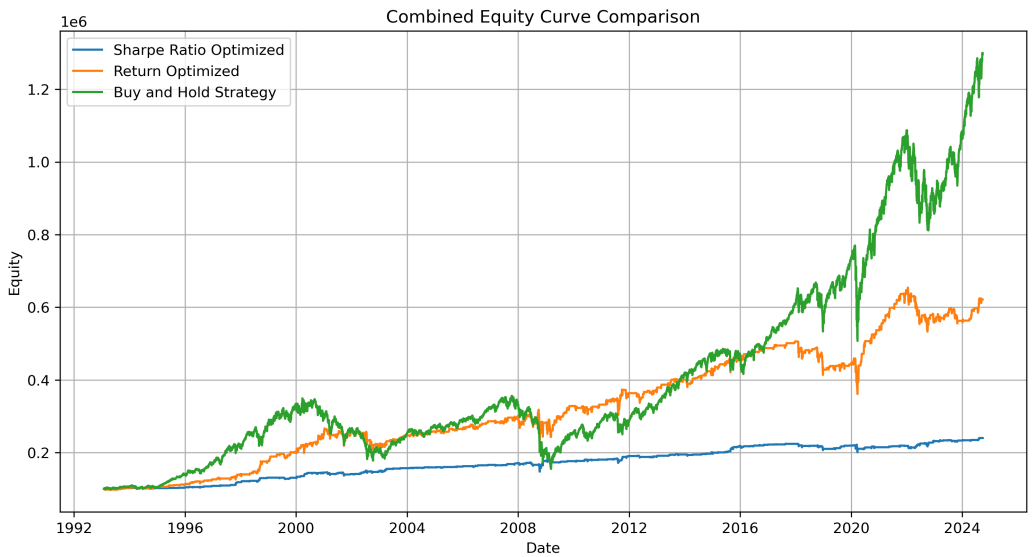

Figure 3.3: Optimized Equity Curve Analysis

The Combined Equity Curve Comparison plot provides a visual representation of the performance of our optimized strategies against a buy-and-hold approach over a 31-year period from 1993 to 2024. This graph offers valuable insights into the long-term behavior and effectiveness of each strategy.

The buy-and-hold strategy, represented by the green line, shows the highest overall return but also exhibits significant volatility. It demonstrates strong growth, particularly in the latter half of the period, but also shows substantial drawdowns, most notably during the 2008 financial crisis and the 2020 COVID-19 market crash.

The Return Optimized strategy, depicted by the orange line, shows a more moderate growth trajectory compared to the buy-and-hold approach. While it doesn't match the peak returns of buy-and-hold, it demonstrates smoother growth with reduced volatility. This strategy appears to mitigate some of the major drawdowns seen in the buy-and-hold approach, particularly evident during market downturns.

The Sharpe Ratio Optimized strategy, represented by the blue line, exhibits the most conservative growth pattern. It shows the lowest overall return but also the least volatility among the three strategies. This strategy maintains a steady upward trajectory with minimal drawdowns, reflecting its focus on risk-adjusted returns rather than absolute performance.

These visual results align with our earlier numerical findings, illustrating the trade-offs between risk and return in each strategy. The plot effectively demonstrates how optimization for different objectives (maximum return vs. risk-adjusted performance) can lead to significantly different long-term outcomes in trading strategy performance.

3.3 Comparison of Original and Optimized Strategies

Comparison of Optimized Strategies with Backtesting.py and Buy and Hold
Performance Metrics for Sharpe, Return Optimizations, and Benchmark Strategies

Metric	Optimized Sharpe	Optimized Return	Backtesting.py	Buy and Hold
Annualized Return	0.028053	0.059404	0.063886	0.084333
Max Drawdown	-0.147444	-0.286117	-0.237414	-0.564737
Sharpe Ratio	0.180878	0.365710	0.458419	0.420198
Volatility	0.049181	0.124302	0.103113	0.186891
Number of Trades	285.000000	484.000000	349.000000	NA
Win Rate	0.708772	0.675620	0.724928	NA
Avg Gain per Trade	4.143358	7.867240	6.600000	NA

3.4 Discussion of Optimization Impact

The optimization process has significantly enhanced the original Rubber Band Trading Strategy. Both optimized versions offer distinct advantages, catering to different investment objectives. The Sharpe Ratio optimized strategy dramatically reduces volatility to 4.92% from the original 10.31%, minimizing risk with a lower maximum drawdown of -14.74% compared to -23.74%.

The Return optimized strategy achieves a solid annualized return of 5.94%, close to the original 6.39%, but with improved risk metrics. Its Sharpe ratio of 0.366 surpasses the original's 0.458, indicating better risk-adjusted returns.

Both optimized strategies show higher win rates and average gains per trade than the original. The Sharpe optimized version has a 70.88% win rate, while the Return optimized achieves 67.56%, both improvements over the original 72.49%.

While the Buy and Hold strategy outperforms in raw annualized returns at 8.43%, its high volatility (18.69%) and maximum drawdown (-56.47%) highlight the value of our optimized strategies in risk management. The optimized versions offer smoother equity curves and controlled risk exposure, appealing to risk-averse investors or those seeking stable returns.

These results underscore the importance of parameter optimization in trading strategy development. I've created versions that can suit different investor profiles – one focused on capital preservation and steady growth, and another aimed at maximizing returns while maintaining better risk control than a buy-and-hold approach.

4 Conclusion

In this study, I have thoroughly examined the Rubber Band Trading Strategy, from its initial implementation to optimized versions. I successfully recreated the original strategy, validating its performance claims. Through optimization, I developed two enhanced versions: one maximizing the Sharpe ratio for better risk-adjusted returns, and another focusing on total returns. Both optimized strategies demonstrated improved performance metrics compared to the original, with lower drawdowns and better risk management.

While neither strategy outperformed a simple buy-and-hold approach in terms of total returns, they showed significantly reduced volatility and drawdowns. This makes them attractive options for risk-averse investors or those seeking more stable returns.